# Listen Before You Classify: A DSP Wake-Up Front End for Energy-Efficient and Reliable Sound Alerts on Wearables

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***Abstract*—Wearable sound-awareness devices for Deaf and hard-of-hearing people run a neural audio classifier on every half second of microphone input, which drains the battery, and pretrained classifiers raise false alarms on safety classes. We place a classical signal-processing front end before the classifier (YAMNet): a high-pass filter, short-time band energies and a per-band minimum-tracking noise floor wake the classifier only when a new sound starts, and a Goertzel resonator bank with a temporal-pattern matcher recognises standard alarm tones directly. The front end costs 0.5 % of one classifier inference. On 8 h of test streams built from real events and real backgrounds in six environments never used for tuning, it lets the classifier sleep 86 % of the time while losing one of 288 detections, while fire-alarm recall on standard alarm tones rises from 29 % to 93 %. A diagnosis of the classifier's false alarms (87 % are sirens labelled as fire alarms) leads to a veto that removes 83 % of them. On naturally recorded audio, the gate was awake for 97 % of 2 765 labelled events in 692 real home recordings and saves about 67 % of classifier compute over a real household's day (99 % in an empty home, 23 % with the TV on), with no false alert in 5.5 h. On held-out real alarm recordings the DSP detector adds 5–13 points of smoke-alarm recall. We also show that synthetic, standard-compliant alarms misjudge both detectors, and characterise real alarms to explain why.**

***Index Terms*—sound event detection, wake-up detector, Goertzel algorithm, noise-floor tracking, energy efficiency, assistive technology**

## I. Introduction

Wearable sound-awareness devices turn important environmental sounds, such as smoke alarms, sirens and door knocks, into vibration and visual alerts for Deaf and hard-of-hearing (DHH) people [1]. They typically run a pretrained neural classifier such as YAMNet [2] on every half second of audio. This has two costs. *Energy*: the classifier runs continuously although nothing new happens most of the time, and battery drain was a key constraint for on-watch sound recognition [1]. *Reliability*: a general-purpose classifier confuses related safety classes and can miss standardised signals such as the ISO 8201 temporal-three (T-3) smoke-alarm pattern [3].

Both problems have a classical signal-processing answer. Most of the day the acoustic scene is stationary, which a cheap onset detector can tell without a neural network; and alarms are *designed* signals, with a known pitch and rhythm that a matched detector can recognise. We ask: *can an adaptive DSP front end that wakes the classifier only when a new sound starts, and that detects alarm tones itself, cut classifier computation substantially without losing detections, in acoustic environments it was never tuned on?*

Our contributions are:

1) A training-free front end (Butterworth high-pass, short-time band energies, minimum-tracking noise floor, Goertzel resonator bank, T-3 matcher) costing 0.5 % of one classifier inference.
2) A controlled comparison against continuous inference, duty cycling and a fixed energy gate, calibrated on 11 environments and tested on 6 unseen ones: 86 % less classifier compute with one of 288 detections lost (64 % with none lost), with a classifier duty that is far more stable across rooms than a fixed threshold.
3) A false-alarm diagnosis of the classifier and a one-line veto that removes 83 % of false alarms.
4) Validation on naturally recorded audio: 97 % of 2 765 real labelled events wake the gate, the compute saving over a real household's day is about 67 %, and held-out real alarm recordings show where the DSP detector helps (in noise) and where it cannot (bells, horns, non-T-3 alarms).
5) Evidence that synthetic, standard-compliant alarms overstate a DSP detector (90 % vs. 25–30 % on real recordings) and understate the classifier, with a characterisation of real alarms that explains the gap. All data are public and the study is reproducible with one command.

## II. Related Work

**Sound awareness for DHH users.** SoundWatch [1] classified environmental sounds on a smartwatch and compared watch-only, phone and cloud architectures, finding battery drain a key concern; ProtoSound [4] personalised recognition from few examples. Neither reduces *when* the classifier runs.

**Wake-up detectors and cascades.** Always-on systems commonly place a low-power detector before a costly stage [5]; keyword spotting wakes large models with small ones [6], and a recent cascade screens audio with a linear detector before a trimmed YAMNet for violence detection [7]. We use interpretable, training-free DSP gates whose threshold follows the room through noise-floor tracking, evaluate them on naturally recorded home audio, and add a second, decision-making DSP path for alarm tones.

**Noise estimation and detection.** Our floor tracker is a simplified form of minimum statistics [8]; the ordered-statistic

CFAR threshold we test as an ablation comes from radar detection [9]. Sigtia et al. compared environmental sound classifiers by accuracy and computational cost, including smoke alarms recorded in homes [10].

## III. Method

### A. Signals and classifier

Audio is sampled at $f_s = 16\,\text{kHz}$, the classifier's native rate. The Nyquist frequency of 8 kHz covers the piezo smoke-alarm band (about 2.8–3.6 kHz) and the 520 Hz low-frequency alarm [11] with its odd harmonics up to the 13th. The classifier, YAMNet (MobileNetV1, 521 AudioSet classes), sees a 0.975 s window every 0.5 s and costs $M_Y = 70\,744\,064$ multiply-accumulates (MACs) per window, counted from its convolution and dense layers. Its scores are mapped to three alert categories (fire alarm, siren, door) by the maximum over category labels. An alert fires when the category score reaches 0.30 in $N$ classified windows in a row ($N = 2$ for fire alarm and siren, 1 for door; a run survives skipped windows if classified windows are at most 2 s apart), with a 20 s cooldown and the most urgent category winning.

### B. Gate path

**High-pass filter.** Butterworth, order 2, $f_c = 100\,\text{Hz}$ ($-3\,\text{dB}$), one biquad

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{1}$$

with a double zero at $z = 1$ and two poles inside the unit circle (Fig. 3). It removes DC, rumble and handling noise below the lowest target component, so they cannot wake the classifier.

**Band energies and noise floor.** Each 20 ms frame ($N = 320$, Hann) is transformed with a 512-point FFT and its power summed in $B = 16$ log-spaced bands (100 Hz–7.6 kHz), smoothed with a one-pole low-pass ($\tau = 60\,\text{ms}$). With $E_b[t]$ the band level in dB, each band's floor follows the minimum with a bounded rise,

$$F_b[t] = \min\big(E_b[t],\, F_b[t-1] + \rho T\big), \qquad \rho = 3\ \text{dB/s}, \tag{2}$$

a simplified form of minimum statistics [8]: it drops to a quieter background at once and absorbs a steady sound within seconds.

**Wake-up.** The novelty $\nu[t] = \max_b(E_b[t] - F_b[t])$ is the largest rise of any band above its own floor, so the threshold is relative to the room. Before each classifier window, if the maximum of $\nu$ over the 0.5 s just received exceeds $\Delta$, the classifier runs, and stays awake for a hangover $H$. The window spans the last 0.975 s, so the onset that woke the gate lies inside the first classified window: waking adds no delay.

### C. Alarm path

**Goertzel resonator bank.** The power of DFT bin $k$ in a frame is computed by the Goertzel recursion [12]

$$s[n] = x[n] + 2\cos(\omega_k)\, s[n-1] - s[n-2], \tag{3}$$

$$|X_k|^2 = s_1^2 + s_2^2 - 2\cos(\omega_k)\, s_1 s_2, \tag{4}$$

a second-order IIR resonator with poles at $e^{\pm j\omega_k}$ on the unit circle, read once per frame. It needs one multiply per sample per probe, against $N \log_2 N$ for a full FFT, which pays off when few frequencies matter. We use 17 probes from 2.8 to 3.6 kHz (50 Hz apart, one bin each) and 520, 1560 and 2600 Hz.

**Tone track and pattern.** A frame is "on" when the strongest probe holds more than a share $r$ of the frame energy (35 % for the three square-wave probes together) and is within 8 dB of its 1.5 s running peak, which rejects reverberant tails. After closing 100 ms dropouts, a fire alarm is declared on three pulses of 0.3–0.8 s separated by gaps of 0.3–0.8 s, the T-3 timing with tolerance. The first design (v3) used $r = 25\,\%$ and also required the 1.5 s pause after the third pulse ("strict"). The final design (v4) was selected on real recordings (Sec. IV-C): $r = 15\,\%$ without the pause check. Its alerts are merged with the classifier's under the same cooldown.

**Siren veto.** A fire alarm from the classifier is accepted only when its score exceeds the siren score in the same window (motivated in Sec. V-C).

### D. Front ends and cost

F0 runs the classifier on every window; F1 on every $k$-th window; F2 wakes it when the high-passed level exceeds a fixed $T$ dBFS; F3 is the adaptive gate; F4, the proposed system, is F3 plus the alarm path and the veto. Per 0.5 s hop the gate costs 194 535 MACs and the alarm path 177 500, counted per stage, together 0.53 % of one classifier inference. Compute relative to F0 is $(d\,M_Y + M_{\text{gate}} + M_{\text{alarm}})/M_Y$, with $d$ the classifier duty (fraction of windows classified).

## IV. Experimental Setup

### A. Test streams

Following the event-mixing protocol of DCASE 2017 Task 2 [13], we build 5-minute streams from DEMAND [14] recordings (8 microphones per environment as separate streams, background level $-50$ to $-35\,\text{dBFS}$). Each event stream holds 6 targets (2 per category) at an SNR from $\{-5, 0, 5, 10, 20\}$ dB over the event's active frames, and 6 distractors (0–20 dB) that never overlap a target; each has an *idle* twin with the background only. Sirens, doorbells and knocks are real recordings from FSD50K [15] and ESC-50 [16] (official splits); distractors are other FSD50K/ESC-50 sounds, half of them easily confused with targets (bells, phones, doors, impacts, beeps, speech, music). Fire alarms are synthesised to ISO 8201 (piezo 2.9–3.5 kHz or a 520 Hz square wave, jittered timing and pitch, a random room response), because no public set of real smoke-alarm recordings exists. Tuning uses 11 environments (7.3 h of event and 7.3 h of idle streams, 528 targets); all reported numbers use 6 others (one per DEMAND category: living room, office, cafeteria, street, bus, park; 4.0 h + 4.0 h, 288 targets).

### B. Calibration

For each gate we choose, on the tuning streams, the lowest-duty setting whose *gate-caused* miss rate (the share of F0's

detections the gated system loses) stays within a budget of 1, 3 or 5 %. Four gate statistics were compared on tuning data (Table II); the selection rule kept the simplest.

### C. Real recordings

**Alarms.** 161 Freesound [17] recordings (CC0/CC-BY/CC-BY-NC): 38 smoke alarms, 30 fire alarms (bells, horns, beeps) and 93 look-alikes that should not raise a fire alarm (microwave, oven and kitchen timers, reversing beepers, alarm clocks, monitors, phones). They are split *by uploader* into a design half and a held-out half, so no recording device or room appears in both. The detector redesign searched probe band (4 widths), tone share (3 values) and pattern rule (T-3, strict T-3, periodic beeps, combinations) on the design half, with a rule fixed in advance: highest alarm recall with no fire alarm on design-half look-alikes. Each clip is tested clean and mixed into a test-environment background at 10 and 0 dB, after 5 s of that background so the noise floor has settled, as on an always-listening device.

**Real scenes.** DESED public evaluation set [18]: 692 real 10 s home recordings with time-stamped labels of 10 classes (2 765 events). SINS [19], [20]: real recordings of one home labelled with the household's activity; we score 1 981 10 s segments (5.5 h, up to 400 per activity), each after a 2 s warm-up. Nothing is tuned on these data.

### D. Metrics

A target is detected if an alert of its category fires between its onset and 1 s after its end. We report recall with 95 % bootstrap intervals that resample whole streams (events in one stream share a background), false alarms per hour, median latency, classifier duty and relative compute; paired differences use the exact McNemar test.

## V. Results

### A. Compute and recall

Table I gives the test results. Duty cycling (F1) cannot save anything: classifying every second window already loses more than 5 % of detections on tuning data, so calibration returns $k = 1$. Both gates keep F0's recall within one or two of 288 detections. At the 5 % budget the adaptive gate F3 lets the classifier sleep 86 % of the time in background-only audio and 71 % in event streams, against 64 % and 52 % for the fixed gate F2, at the same recall; at the 3 % budget the proposed system loses no detection at all with 64 % less compute. Fig. 1 shows the full trade-off: the adaptive gate dominates the fixed one at low duty, and duty cycling collapses.

### B. Robustness across rooms

A fixed level threshold depends on how loud a room is. In background-only audio its duty at the 5 % budget ranges from 0.7 % (bus) to 69 % (cafeteria), while the adaptive gate stays between 1.4 % (street) and 24 % (bus) in all six unseen rooms (Fig. 2), because its threshold is relative to each band's own floor.

TABLE I
Test results (6 unseen environments, 288 targets). FA/h on event streams; duty in background-only streams; compute relative to F0.

| Front end | Recall (%) | Fire (%) | Siren (%) | Door (%) | FA/h | Duty (%) | Comp. |
|---|---|---|---|---|---|---|---|
| F0 always on | 32.6 | 29.2 | 53.1 | 15.6 | 11.5 | 100 | 1.00 |
| F0 + veto | 32.6 | 29.2 | 53.1 | 15.6 | 2.0 | 100 | 1.00 |
| F0 + alarm DSP + veto | 53.8 | 92.7 | 53.1 | 15.6 | 2.0 | 100 | 1.00 |
| F1 every $k$-th | no $k > 1$ meets even the 5 % budget | | | | | | |
| F2 fixed gate (5 %) | 31.9 | 28.1 | 53.1 | 14.6 | 10.5 | 36 | 0.36 |
| F3 adaptive gate (5 %) | 32.3 | 29.2 | 52.1 | 15.6 | 11.3 | 14 | 0.14 |
| F4 (1 % budget) | 53.8 | 92.7 | 53.1 | 15.6 | 2.0 | 66 | 0.67 |
| F4 (3 % budget) | 53.8 | 92.7 | 53.1 | 15.6 | 2.0 | 35 | 0.36 |
| **F4 proposed (5 %)** | **53.5** | **92.7** | 52.1 | 15.6 | **2.0** | **14** | **0.14** |
| F4 with DSP v3 (5 %) | 52.1 | 88.5 | 52.1 | 15.6 | 2.0 | 14 | 0.14 |

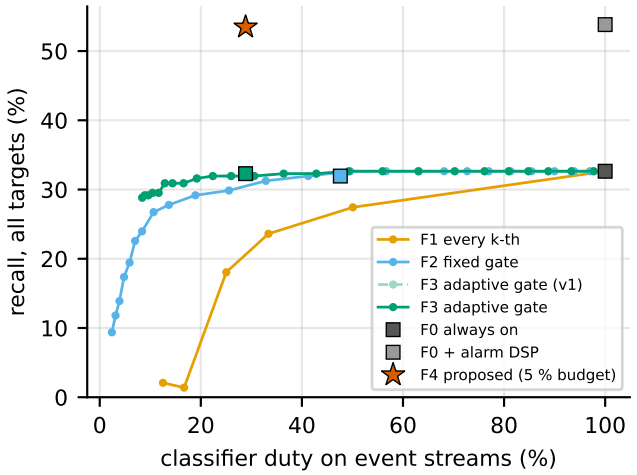



Fig. 1. Recall against classifier duty on the test event streams for the duty-cycled (F1), fixed (F2) and adaptive (F3) gates; markers: settings chosen on tuning data.

**Gate statistic ablation.** Longer smoothing with a two-band rule, and an ordered-statistic CFAR threshold (median novelty of the last 10 s) [9], did not improve on the simple statistic (Table II). CFAR was clearly worse: during a long sound such as a siren its reference median rises, and the statistic sinks just when the classifier's scores peak.

### C. False alarms

All of F0's false alarms occur in event streams (none in 4 h of background): 40 of 46 are fire alarms raised during sirens, because YAMNet also scores *Alarm* on sirens and the fire alarm has priority. Tuning the classifier's thresholds on the tuning streams (best 0.275 instead of 0.30) does not change recall or false alarms. The siren veto cuts false alarms from 11.5 to 2.0 per hour (83 %) at unchanged recall and lowers the median alert latency from 1.85 to 1.55 s. The DSP alarm path adds no false alarm on the 1 000+ everyday distractors.

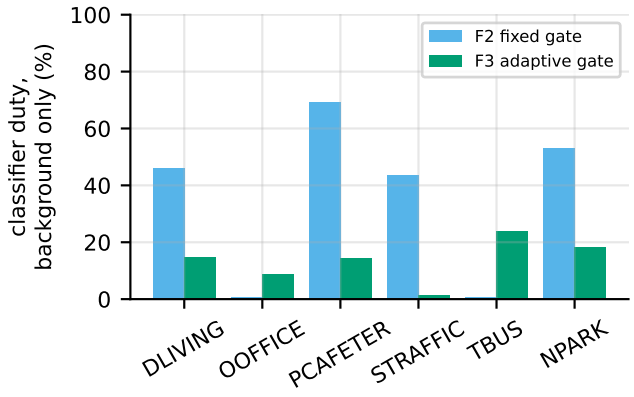



Fig. 2. Classifier duty in background-only audio per unseen test environment.

TABLE II
CLASSIFIER DUTY (%) ON THE TUNING ENVIRONMENTS WITHIN EACH GATE-CAUSED MISS BUDGET.

| Gate | 1 % | 3 % | 5 % |
|---|---|---|---|
| F1 every $k$-th | 100 | 100 | 100 |
| F2 fixed level | 46.7 | 37.8 | 30.7 |
| F3 max band rise (selected) | 49.6 | 31.0 | 21.4 |
| + 150 ms smoothing, two bands | 50.6 | 28.6 | 23.1 |
| + OS-CFAR | 85.2 | 47.1 | 36.1 |
| + both | – | 45.3 | 35.4 |

### D. Alarm detection: synthetic versus real

On the synthetic ISO 8201 alarms of the test streams, the DSP path raises fire-alarm recall from 29 % to 93 % ($p < 10^{-15}$); the classifier never detects the 520 Hz alarm, while the DSP path detects most of them (85 % in the first design). The gate does not delay alerts (median 1.84 s vs. 1.85 s for F0); the alarm path needs about 2.4 s to observe three pulses.

Real recordings tell a different story. Characterising the 32 design-half real alarms with DSP (FFT peak of tonal frames, beep timing) shows a median pitch of 3.0 kHz but only 54 % within the 2.8–3.6 kHz probes (range 1.0–3.5 kHz), and only 6 of 32 follow the T-3 pattern: 4 are continuous tones, 3 other rhythms, and 19 are not narrowband tones at all (bells, horns, distorted recordings). A tone-and-pattern detector can therefore only reach a minority of real alarms. The redesign raised design-half recall from 18 % to 25 % with no look-alike alarm; wider bands or a periodic-beep rule reached 27–33 % but fired on 2–17 % of look-alikes and were rejected.

On held-out real recordings (Table IV) the classifier is much stronger than on synthetic alarms (43–62 % clean vs. 29 %) and the DSP detector much weaker (26–30 % vs. 93 %). Combined, the proposed system finds 5–13 points more smoke alarms than the always-on classifier, most at 0 dB, where the classifier degrades and the narrowband detector does not. The gate itself was awake during 98/91/78 % of the smoke alarms. It costs one of 13 fire bells at 10 and 0 dB, where quiet bells do not always wake it. The DSP detector fired on 1 of 50 held-out look-alikes, against 6 for the classifier.

TABLE III
STRESS TEST (20 CLIPS EACH, 5 DB SNR): SHARE RAISING A FIRE ALARM.

| Signal | Fire? | YAMNet | DSP strict | DSP v4 |
|---|---|---|---|---|
| T-3, 3.1 kHz piezo (in spec) | yes | 75 | 85 | 85 |
| T-3, 520 Hz square (in spec) | yes | 0 | 90 | 90 |
| T-3, 2.6 kHz / 4.0 kHz | yes | 70 / 65 | 50 / 0 | 50 / 0 |
| T-3, timing $\times 0.7$ / $\times 1.5$ | yes | 90 / 70 | 35 / 35 | 35 / 30 |
| T-3, reverberation 1.5 s | yes | 60 | 75 | 70 |
| T-4 carbon-monoxide pattern | no | 50 | 0 | 0 |
| Continuous 1 Hz beeper, 3.1 kHz | no | 80 | 0 | 85 |
| Single triple beep, 2.8 kHz | no | 15 | 0 | 0 |

TABLE IV
HELD-OUT REAL RECORDINGS (FREESOUND, UNSEEN UPLOADERS): SHARE RAISING A FIRE ALARM (%), CLEAN / 10 DB / 0 DB. F4 IS THE FULL GATED SYSTEM.

| Group ($n$) | YAMNet (F0) | DSP v4 alone | F4 proposed |
|---|---|---|---|
| Smoke alarms (23) | 43 / 43 / 22 | 30 / 26 / 26 | **52 / 48 / 35** |
| Fire bells, horns (13) | 62 / 62 / 31 | 0 / 0 / 0 | 62 / 54 / 23 |
| Look-alikes (50) ↓ | 12 / 6 / 4 | 2 / 2 / 0 | 14 / 8 / 4 |

### E. Naturally recorded audio

On DESED's 692 real home recordings the gate was awake for 96.9 % of all labelled events (alarm/bell/ringing 99.0 %, speech 98.9 %, dishes 95.1 %, vacuum cleaner 87.5 %, the lowest, as a steady noise is partly absorbed by the floor), and the gated system kept 30 of the 32 alerts of the always-on classifier. Table V gives the classifier duty over real home life. It follows the household: under 1 % in an empty home, 10 % while working, 77 % with the TV on. Weighted by how SINS divides the day, the classifier runs 33 % of the time, a 67 % compute saving. Neither the classifier (with the veto) nor the DSP path raised an alert in the 5.5 h.

## VI. DISCUSSION

**What the front end buys.** The main gain is compute with reliability: the same detections at a fraction of the classifier's compute, fewer false alarms, and no added delay from waking. The saving depends on the scene and should be reported per activity, not as one number: our mixed scenes suggest 86 %, a real household 67 %, an empty home 99 %, a home with the TV on 23 %.

**Where DSP detection helps, and where it cannot.** A matched DSP detector is robust in noise and rarely fooled by look-alikes, but it only reaches alarms that are the narrowband patterns it encodes. On real recordings that is a minority, so it is a safety net beside the classifier, not a replacement. Synthetic, standard-compliant test alarms hide this: they made the DSP detector look three times better and the classifier half as good as on real recordings. Detector designs should be validated on real recordings, split by recording source.

**Limitations.** Real smoke-alarm recordings are scarce: 38 in total, so the held-out differences (2–3 clips) are not statistically significant. SINS segments were scored independently after a

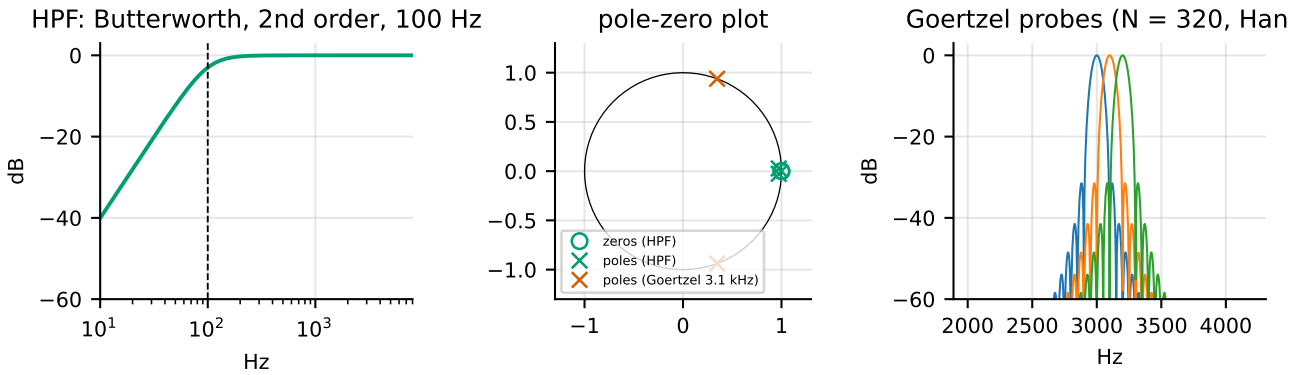



Fig. 3. DSP design: high-pass filter magnitude response, pole-zero plot (HPF and one Goertzel resonator) and Goertzel probe responses ($N = 320$, Hann).

TABLE V
REAL HOME LIFE (SINS, 5.5 H): CLASSIFIER DUTY OF THE PROPOSED SYSTEM PER ACTIVITY.

| Activity | h | Duty | Activity | h | Duty |
|---|---|---|---|---|---|
| Absence | 1.1 | 0.9 % | Eating | 0.3 | 43.6 % |
| Vacuuming | 0.1 | 7.4 % | Dishwashing | 0.2 | 50.6 % |
| Working | 1.1 | 10.2 % | Visitors, calls | 0.6 | 74.5 % |
| Cooking | 0.7 | 28.1 % | Watching TV | 1.1 | 76.7 % |
| Weighted by the full SINS day: 33 % duty, 67 % saving | | | | | |

short warm-up, not as continuous episodes. We count multiply-accumulates, not battery energy; the microphone and the rest of the device keep their own cost. The rule without the pause check (v4) fires on continuous 1 Hz beepers (Table III); the strict rule does not; the first, strict design reached 7 points less recall on the design-half real alarms, and the choice depends on the deployment. Fire bells and horns are outside the detector's design. Results are for one classifier.

## VII. CONCLUSION

A classical front end costing 0.5 % of one neural inference lets a wearable's sound classifier sleep most of the time without losing detections, adapts to unseen rooms where a fixed threshold does not, and, with a matched alarm detector and a siren veto, raises alarm recall and removes most false alarms. Validation on naturally recorded audio confirmed the gate and a 67 % compute saving over a real household's day, and showed that real alarms, unlike synthetic ones, are mostly beyond a tone-and-pattern detector. Future work: continuous real-home recordings with labelled safety events, on-device energy measurement, and detectors for the alarm types characterised here.

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